

YASHESH VAIDYA

MSc Applied Cyber Security, Queen's University Belfast | Cyber Security | Forensics Analyst | 3+ YoE

| Belfast, UK | | [Gmail](#) | [LinkedIn](#) | [Github](#)

PROFESSIONAL SUMMARY

Level 7 MSc in Applied Cyber Security candidate at Queen's University Belfast, holding **Certified Ethical Hacker (CEH v13)**, **CEH Practical**, and **Computer Hacking Forensic Investigator (CHFI v11)** certifications. Hands-on experience in digital forensics, **incident response**, security investigations, **vulnerability assessment**, and network security. Developed academic projects in **threat intelligence**, secure software development, and **AI/ML security** using Python, with practical experience in threat analysis, log analysis, **chain-of-custody** evidence handling, and **e-discovery**. Familiar with MITRE ATT&CK and **MITRE CWE** frameworks, **DevSecOps** practices, and security operations principles.

SKILLS

- **Functional Competencies:** Vulnerability Assessment | IP Scanning | System Log Analysis | Computer Forensics | Mobile Forensics | Email Forensics | Network Forensics | Penetration Testing | Digital Evidence Collection | Chain of Custody | Incident Response | Threat Modelling | MITRE ATT&CK Framework | Network Security Monitoring | Security Auditing | Threat Hunting | OSINT | Artifact Analysis | Secure Coding | Technical Reporting | Risk Assessment | E-Discovery | Problem Solving | Cross-Functional Collaboration.
- **Technical Tools:** Kali Linux | NMAP | Nessus | Wireshark | Snort | Metasploit | Burp Suite | Volatility | FTK Imager | Autopsy | Cellebrite UFED | Oxygen Forensics | OSForensics | Agent Ransack | Multi Memu | Android Backup-Trans | Steganography | Email Header Tracer | Bandit (SAST) | defusedxml | Git | GitLab | Microsoft Office 365
- **Programming Languages:** Python | C++ | Bash / Shell Scripting | SQL
- **Web Development:** HTML | CSS | Bootstrap | JavaScript | PHP | Figma (UI/UX)
- **Other Skills:** Graphic Designing (Canva, Adobe XD, Photoshop) | AI Tools (Claude, ChatGPT, Codex)

WORK EXPERIENCE

Digital Forensic Analyst | Freelancer

May 2025 – Aug 2025

- Conducted independent **digital forensic investigations** for clients, performing **Forensic Acquisition**, **Forensic Imaging**, and **Chain of Custody** documentation using **FTK Imager** and **Autopsy** to preserve **Data Integrity**.
- Performed **Mobile Forensics** and **Email Forensics** using **Oxygen Forensics** and email header analysis, recovering **Deleted Data** and correlating artifacts to build **investigation timelines**.
- Prepared detailed **Forensic Reports** and evidentiary summaries, translating technical findings into clear, actionable insights to support client decision-making, **Incident Response**, and **legal proceedings**.

Guest Faculty – Cybersecurity & Digital Forensic | Rizvi College of Engineering

Feb 2025 – Apr 2025

- Developed and delivered an Honours course in **Cybersecurity**, **Digital Forensics**, and **Application Security**, covering **Secure SDLC**, **Threat Modelling**, **Vulnerability Assessment**, **Digital Evidence Handling**, and **Forensic Investigation Methodologies**.
- Conducted hands-on training using **FTK Imager**, **Oxygen Forensic Suite**, **Nmap**, and **Nessus** for **Forensic Imaging**, **Evidence Collection**, **Artifact Analysis**, **Security Assessment**, and **System Hardening** activities.
- Designed and evaluated practical labs on **Cybersecurity** and **Digital Forensics**, enabling students to apply **Incident Investigation**, **Evidence Preservation**, and **Security Analysis** techniques in controlled environments.

Forensics Analyst | Covert Infolabs Private Limited

Dec 2024 – Feb 2025

- Collaborated with **Law Enforcement Agencies** during investigations and raids, performing **Incident Triage**, **Digital Evidence Identification**, **Evidence Preservation**, and **Forensic Analysis** in accordance with legal and regulatory requirements.
- Conducted **Forensic Acquisition**, **Forensic Imaging**, and **Chain of Custody** documentation using industry-standard tools and write-blockers, ensuring **Data Integrity** and admissibility of evidence in legal proceedings.
 - Recovered **Deleted**, **Encrypted**, and **Hidden Data**, and prepared detailed **Forensic Reports**, **Investigation Documentation**, and **Evidentiary Records** to support **Fraud Investigations**, regulatory actions, and legal proceedings.

Forensics Intern | Cybermars Forensic Private Limited

Nov 2023 – Dec 2024

- Assisted tax authorities by analysing seized digital data, performing e-discovery and forensic reviews, and uncovering key discrepancies during **regulatory compliance** investigations.
- Examined IT infrastructures through **network forensics** and **artifact analysis** of system logs, registry entries, and file system data to identify security incidents, build **investigation timelines**, and secure actionable evidence.
- Performed mobile device forensics using **Cellebrite UFED** and **Oxygen Forensics** to extract, decode, and analyse data from smartphones and tablets, contributing to **investigation timelines** and evidentiary reporting.

EDUCATION

Queen's University Belfast, Belfast, Northern Ireland, UK

Sep 2025 – Sep 2026

Master of Science in Applied Cyber Security

Subjects: Foundation of Cyber Security, Pen-testing & Ethical Hacking, Network Security and Monitoring, Software Assurance, Cyber-AI, Cryptography, Data Privacy and the Law.

Rajiv Gandhi Institute of Technology, Mumbai, India

May 2024

Bachelor of Engineering in Electronics & Telecommunications, With Honours

(CGPI 7.61/10)

Specialisation in AI & Machine Learning

Subjects: IT Infrastructure & Security, Computer Communication Networks, Digital Forensics, Cloud Computing and Security, Cyber Security and Laws, Network Management in Telecommunications.

PROJECT EXPERIENCE

Snort Community Ruleset Threat Coverage Analysis | Python, MITRE ATT&CK, LLMs, GitLab – ([Github](#)) MSc

Dissertation Project – Queen's University Belfast

- Built a five-stage automated **Python** framework that parsed and analysed 4,017 **Snort 3 Community rules** across 27 classtype categories, mapping them to the **MITRE ATT&CK** framework validated against ATT&CK STIX 2.1 (v19.1).
- Implemented direct and **heuristic ATT&CK mapping** using rule features (msg, content, **PCRE**, references, service, ports), exposing that only 10.70% of rules map to ATT&CK and identifying 34 rules citing **revoked technique IDs**.
- Integrated **LLM-based classification** (**NVIDIA Nemotron 3 Ultra** 550B via API) to re-predict mapped rules and classify 3,587 unmapped rules, with a manually adjudicated **blind validation** across Claude, ChatGPT and Gemini.
- Delivered evidence-based **coverage analysis** of modern threats (**ransomware**, **DNS abuse**), detection-difficulty tiering, and programmatic **matplotlib** visualisations, with code maintained on **GitLab**.

CWE Vulnerability Analysis System | Python, XML Parsing, GitLab – ([Github](#)) Academic

Software Assurance Project

- Built a **Python**/Flask web application ingesting the **MITRE CWE** XML dataset with XXE-safe parsing (**defusedxml**), delivering 8 REST API endpoints and an interactive dashboard with CIA impact and CVE-ranked views.
- Implemented defence-in-depth **secure coding** with layered input validation and 7 security response headers (CSP, HSTS, X-Frame-Options), plus a high-risk composite view to surface remediation priorities.
- Delivered a 5-stage **DevSecOps CI/CD** pipeline on **GitLab** – lint (**Flake8**), **SAST** (Bandit), dependency scanning (**pip-audit**), testing (**pytest**) and versioned build artefact – following **SSDLC** principles.

FaceGuard AI Security Assessment | Python, PyTorch, Adversarial Machine Learning – ([Github](#)) Academic

Cyber-AI Project

- Conducted a structured **Red Team / Blue Team AI/ML security assessment** of FaceGuard, a CNN-based facial recognition access control system, simulating an attack chain from model theft to privilege escalation.
- Executed a black-box model stealing attack to train a proxy CNN replicating the victim at ~98% accuracy; implemented **FGSM** and **PGD** adversarial attacks and a targeted impersonation attack for privilege escalation.
- Demonstrated ~99% **PGD** transfer success from proxy to victim; applied Madry et al. adversarial training to produce Robust_FaceGuard, evaluating the accuracy-robustness tradeoff for real-world deployment.

COURSES AND CERTIFICATIONS

- **Certified Ethical Hacker (CEH)** by EC-Council | Oct 2025
- Certified Ethical Hacker Practical by EC-Council | Aug 2025
- **Computer Hacking Forensic Investigator (CHFI)** by EC-Council | Dec 2024
- OSForensic Triage Certification (OSFTC) by PassMark Software | Jul 2024
- Certificate in Web Development by IACT Education Pvt Ltd | Nov 2021
- Certificate in Ethical Hacking by IACT Education Pvt Ltd | Oct 2021

EXTRACURRICULAR ACTIVITIES

- Volunteered at **Q-Con** (Jun 2026), supporting a large-scale event attended by approximately 10,000 people.
- Competed in the **Capture the Flag (CTF) Pilot Programme** at CSIT, Queen's University Belfast (Mar 2026).
- Competed in the **Capture the Flag (CTF)** event at **Kainos** (Mar 2026).
- Participated in the **Master Your Leadership Programme**, Queen's Graduate School, QUB (2026), collaborating in a multidisciplinary team to present a community-focused project to industry professionals.
- Led the **IETE** departmental committee at Rajiv Gandhi Institute of Technology, Mumbai as **Chairperson** (2022–2024), progressing from Technical Member to TE Coordinator; led events, workshops, and team management.